

AI Supply Chain & Inventory Intelligence Platform

CBS2003- Information Security Course Project

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ABSTRACT

Efficient inventory and supply chain management are essential for organizations to reduce costs, prevent stock shortages, and respond to changing market demands. This project proposes an Autonomous Supply Chain Control Tower, an AI-powered platform that integrates demand forecasting, inventory optimization, supplier performance analysis, and procurement planning into a single intelligent system. Using historical sales, inventory, and supplier data, the platform predicts future demand, recommends optimal reorder quantities, monitors stock levels, and identifies potential supply chain risks. Interactive dashboards provide real-time insights into inventory health, warehouse operations, and key performance indicators, enabling faster and more informed decision-making. By combining machine learning, optimization techniques, and business analytics, the proposed solution enhances supply chain efficiency, minimizes inventory costs, and improves operational resilience across manufacturing, retail, and logistics organizations.

PROBLEM STATEMENT

Modern organizations face significant challenges in managing inventory and supply chain operations due to fluctuating customer demand, supplier delays, inefficient stock planning, and limited real-time visibility across the supply chain. Traditional inventory management systems primarily focus on recording stock levels but lack intelligent capabilities to predict demand, optimize inventory, and proactively respond to disruptions. These limitations often result in stock shortages, excess inventory, increased operational costs, delayed deliveries, and reduced customer satisfaction.

There is a need for an integrated, data-driven solution that combines demand forecasting, inventory optimization, supplier performance analysis, and real-time monitoring to support informed decision-making. Such a system can help organizations improve supply chain efficiency, reduce costs, enhance operational resilience, and ensure the timely availability of products in a dynamic business environment.

LITERATURE REVIEW

Literature Review 1: Intelligent Supply Chain Management: A Systematic Literature Review on Artificial Intelligence Contributions (Teixeira et al., 2025)

In order to investigate how artificial intelligence affects supply chain management, Teixeira et al. (2025) evaluated 66 papers. According to the study, AI enhances logistics, supplier selection, inventory control, and demand forecasting through data-driven decision-making. It also brought attention to issues like data governance, scalability, and moral dilemmas. The results promote the creation of AI-powered systems that improve supply chain resilience, efficiency, and real-time decision-making—all of which are in line with the project's goals.

Literature Review 2: Resilient and Intelligent Supply Chains: Advances and Challenges in AI-Driven Optimization and Forecasting

This analysis examines how supply chain optimization is enhanced by operations research (OR) and artificial intelligence (AI) using digital twin technologies, inventory planning, and demand forecasting. According to the study, hybrid AI-OR approaches perform better than conventional techniques when handling interruptions, multi-level supply chains, and demand unpredictability. Additionally, it highlights the necessity of strong decision-making, explainable AI, and improved system integration. The creation of intelligent, adaptable supply chain systems, such as the project under consideration, is supported by these discoveries.

Literature Review 3: Examining the Integration of Artificial Intelligence in Supply Chain Management from Industry 4.0 to Industry 6.0

The importance of artificial intelligence in changing supply chain management from Industry 4.0 to Industry 6.0 is examined in this systematic review. According to the study, AI increases supply chain resilience and sustainability while also improving demand forecasting, inventory control, supplier selection, and operational efficiency. It also draws attention to issues including AI integration, labor skill gaps, and cybersecurity threats. The results complement the goals of the proposed project by highlighting the necessity of intelligent, human-centered, and sustainable supply chain systems.

OBJECTIVES

- To develop an AI-powered supply chain management system that predicts demand, optimizes inventory levels, and enhances procurement decisions using real-time and historical data.
- To improve supply chain efficiency and resilience by providing real-time monitoring, supplier performance analysis, and intelligent decision support through predictive analytics and interactive dashboards.

REFERENCES

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